

COMBATING COVID-19

VACCINES

In 2020, a pandemic swept across the globe. It was caused by a harmful microorganism—an organism so small, it can only be seen with a microscope—called a virus. This virus, known as a coronavirus, infected people's lungs and caused a disease called COVID-19. To stop healthy people from catching the virus, scientists developed vaccines. These are medicines that teach the body to identify and fight off the virus. There are similar vaccines for a wide range of diseases.

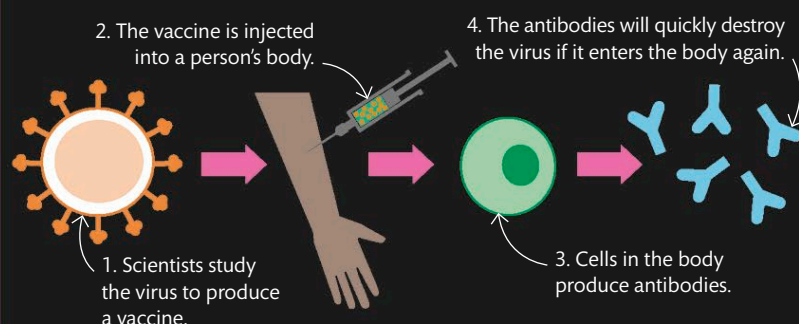
VIRUS VARIATIONS

Viruses keep changing their structure, giving rise to different versions of the same virus called variants. This is why every year a flu vaccine is developed to combat the latest variant infecting people. Every vaccine is tested on volunteers to ensure it is safe and effective.



HOW VACCINES WORK

Vaccines make use of the body's system for fighting off disease, called the immune system. Part of the immune system learns to identify pathogens (harmful viruses or bacteria) that enter the body so that the body can destroy them quickly if they enter it again at a later time. It does this by producing substances called antibodies, which mark the pathogens for destruction whenever they appear. Vaccines cause the immune system to produce antibodies for a specific pathogen without the pathogen being present. This means the body will be protected against a disease without having to be exposed to it first.



The virus has spikes on its surface that make it distinctive to the immune system.

Antibodies produced by the immune system

